Generative AI and Large Language Models (LLMs)

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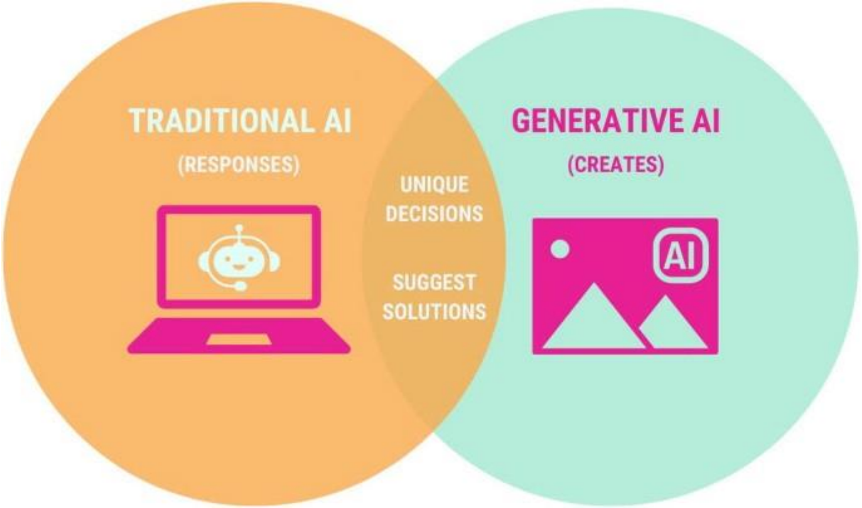
Introduction to Generative AI

What is Generative AI?

- These models generate new content by learning patterns from vast datasets.
- Unlike traditional AI, it generates unique outputs

Why is it important?

- Used in Healthcare, Education, Marketing, Entertainment
- Capable of producing high-quality, human-like content across different formats.



Types of Generative AI Models

- 📝 Text Generation (GPT, Llama, Gemini)
- 🎨 Image Generation (DALL·E, MidJourney)
- 🎥 Video Generation (Runway ML, Pika Labs)
- 🎵 Music & Audio Generation (Jukebox, ElevenLabs)
- 💻 Code Generation (GitHub Copilot, Code Llama)

Focus on Large Language Models (LLMs)

What is an LLM?

- AI that generates human-like text
- Trained on billions of words from books, websites, and code repositories.

Popular LLMs:

- GPT-4 (OpenAI)
- Llama 3 (Meta)
- Google PaLM
- Claude 3

How LLMs Work?

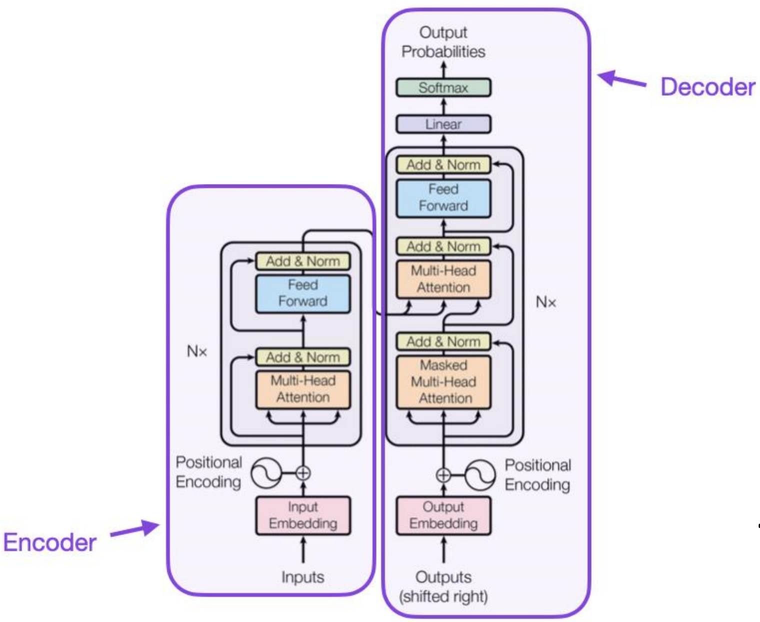


Figure 1: The Transformer - model architecture.

Natural Language Processing (NLP)

- NLP enables LLMs to understand and generate human language.
- Key NLP tasks include:
  - **Tokenization:** Splitting text into words/subwords.
  - **Named Entity Recognition (NER):** Identifying names, dates, locations.
  - **Part-of-Speech Tagging:** Classifying words (noun, verb, adjective).
  - **Semantic Understanding:** Interpreting sentence meaning.

Transformer Architecture

- Uses self-attention to process words in context.
- Unlike RNNs, parallel processing makes it faster and more efficient.
- Assigns importance scores to words based on relevance.

How LLMs Work?

Pretraining & Fine-Tuning

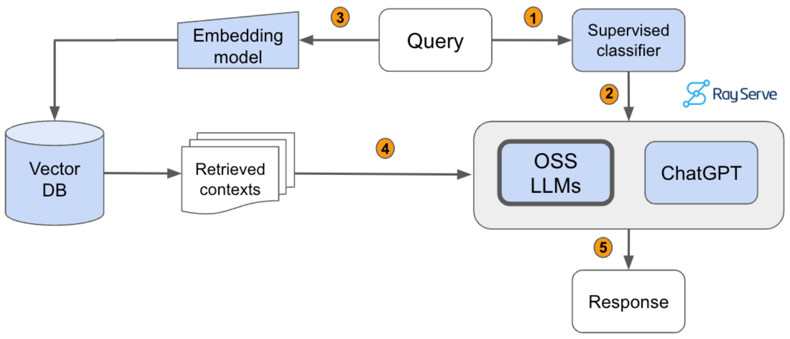
- **Pretraining:** LLMs learn from massive datasets (books, articles, code).
- **Fine-Tuning:** Adjusted for specific applications (e.g., chatbots, medical AI).

Vector Databases & Retrieval (RAG)

- LLMs have limited memory and can't access real-time data.
- Vector Databases (ChromaDB, FAISS, Pinecone) retrieve information before generating responses.
- Retrieval-Augmented Generation (RAG) helps LLMs fetch up-to-date facts.

Text Generation Process

- Uses probability-based techniques:
  - ➢ **Temperature Scaling:** Controls randomness in responses.
  - ➢ **Top-k Sampling:** Limits choices to the most probable words.



Key Features of LLMs

- ✓ **Contextual Understanding**
- ✓ **Multimodal Capabilities**
- ✓ **Code Generation & Analysis**
- ✓ **Task-Specific without Fine-Tuning**
- ✓ **Scalability & Efficiency**

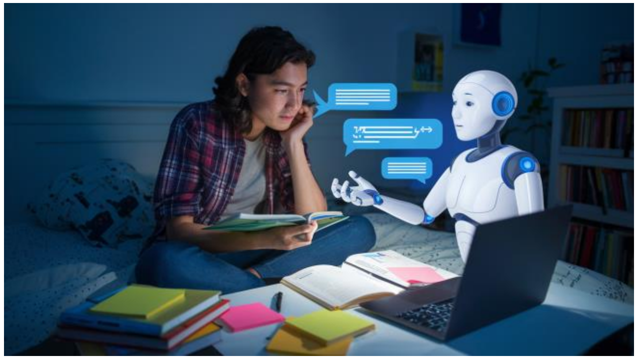
Applications of LLMs

✍️ **Content Creation:** Blogs, Marketing, Social Media

💬 **Customer Support:** AI Chatbots (ChatGPT, Google Assistant)

💻 **Code Generation:** GitHub Copilot, Code Llama

📚 **Education:** AI Tutors, Summarization Tools



Demonstration - Cold Email Generator using Llama 3.3

- **Tech Stack Used:** Llama 3.3, LangChain, ChromaDB, Streamlit, Groq.
- **Purpose:** Automates cold email writing for software service companies.
- **Why Groq?** Ultra-fast inference using LPUs.
- **Process Workflow:**
  - Scrape job listings (Nike example) using LangChain.
  - Extract key details (role, skills, experience) with Llama 3.3.
  - Fetch relevant portfolio links from ChromaDB.
  - Generate personalized cold email using Llama 3.3.

Demonstration - Cold Email Generator using Llama 3.3

Parsing Job Details

```
[7]: from langchain_core.output_parsers import JsonOutputParser

json_parser = JsonOutputParser()
json_res = json_parser.parse(res.content)
json_res
```

```
[7]: {'role': 'Software Engineer II - Consumer Product & Innovation',
 'experience': '2+ years of hands-on industry software development experience',
 'skills': ['Front-end frameworks like React, Angular, or Vue.js',
  'Cloud architecture, modern DevOps, infrastructure as code, CI/CD and related tools',
  'AWS products including Lambda, Step Functions, DynamoDB, Elasticsearch, s3',
  'Modern testing methodologies and frameworks such as Mocha, Jasmine and Jest',
  'Architectural design patterns and computer-science fundamentals',
  'Implementing and integrating AI, Machine Learning and related data solutions'],
 'description': 'We're looking for multiple Software Engineers to solve complex software engineering problems supporting Nike's pursuit of delivering state of the art tools to our Consumer Product & Innovation community.'}
```

Generating Cold Email

```
chain_email = prompt_email | llm
res = chain_email.invoke({"job_description": str(job), "link_list": links})
print(res.content)
```

```
Subject: Expert Software Engineering Solutions for Nike's Consumer Product & Innovation

Dear Hiring Manager,

I came across the job posting for a Software Engineer II - Consumer Product & Innovation at Nike, and I'm excited to introduce KrishV, an AI & Software Consulting company that can help you solve complex software engineering problems. With our expertise in front-end frameworks like React, Angular, and Vue.js, we can support the development of state-of-the-art tools for your Consumer Product & Innovation community.

Our team has hands-on experience with cloud architecture, modern DevOps, infrastructure as code, CI/CD, and related tools, including AWS products like Lambda, Step Functions, DynamoDB, Elasticsearch, and S3. We're well-versed in modern testing methodologies and frameworks such as Mocha, Jasmine, and Jest, ensuring the delivery of high-quality software solutions.

At KrishV, we've empowered numerous enterprises with tailored solutions, fostering scalability, process optimization, cost reduction, and heightened overall efficiency. Our portfolio includes successful projects in:

* Front-end development: https://example.com/typescript-frontend-portfolio, https://example.com/vue-portfolio
* DevOps: https://example.com/devops-portfolio
* Full-stack development: https://example.com/full-stack-js-portfolio
* Machine Learning: https://example.com/ml-python-portfolio

We're confident that our expertise in architectural design patterns, computer-science fundamentals, and AI/ML integration can help Nike achieve its goals. Our team is passionate about delivering innovative solutions that drive business growth and customer satisfaction.

I'd love to schedule a call to discuss how KrishV can support Nike's software engineering needs. Please let me know if you're interested, and we can schedule a time that suits you.

Best regards,

Krishna
Business Development Executive
KrishV
```

Challenges & Limitations of LLMs

⚠ **Bias in AI Models:** May produce biased outputs

💭 **Hallucinations:** Can generate inaccurate information, but techniques like Retrieval-Augmented Generation (RAG) and fact-checking models help mitigate this.

🔒 **Data Privacy Concerns:** LLMs trained on public datasets may inadvertently expose sensitive information.

💰 **Computationally Expensive:** Requires powerful GPUs/TPUs and large-scale infrastructure.

Future of LLMs

🔗 **Multimodal AI:** Text + Images + Video

⚡ **Next-gen LLMs:** Faster, More Efficient, and Optimized for Edge AI.

🔍 **Advancements in RAG:** AI models will rely more on real-time, verified sources for accuracy.“

👨‍💻 **Personalized AI Assistants:** Smarter, user-adaptive AI. Future AI copilots for coding, healthcare, and finance.

Thank You